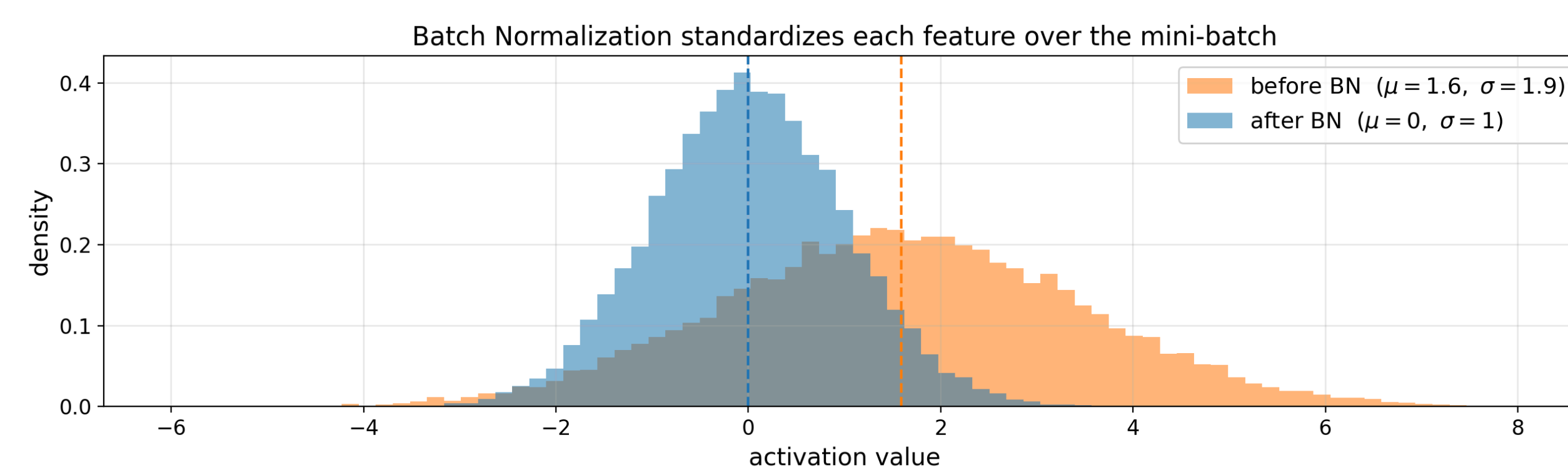


01. Introduction

Batch Normalization (BN) rescales every feature of a mini-batch to mean 0 and variance 1:

$$y = \gamma \cdot (x - \mu_B) / \sigma_B + \beta$$

This simple step lets networks train faster and reach **higher accuracy**, making BN a standard layer in deep learning.



02. Objective

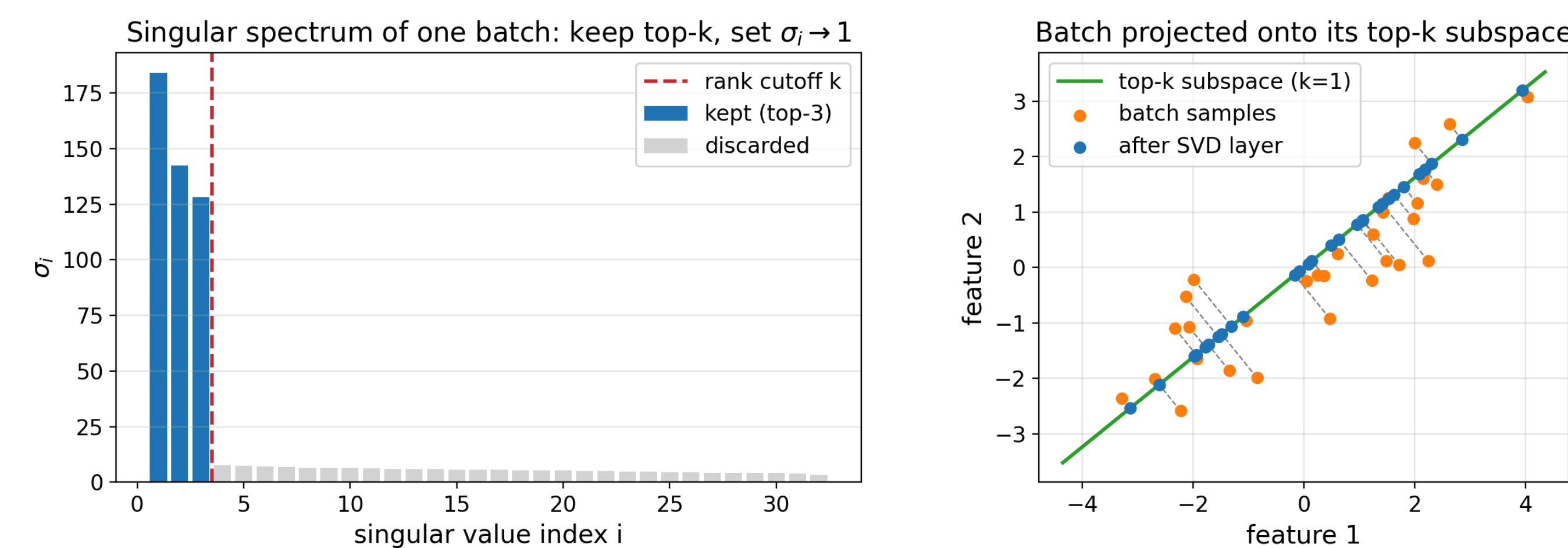
Rank Reduction Autoencoders (Mounayer et al., 2024) regularize networks with the truncated **SVD**, no extra loss needed. Like BN, it **changes the statistics of the batch**.

Can the SVD alone normalize a batch as well as BN?
We build it into a standalone **SVD layer** to find out.

03. Methodology

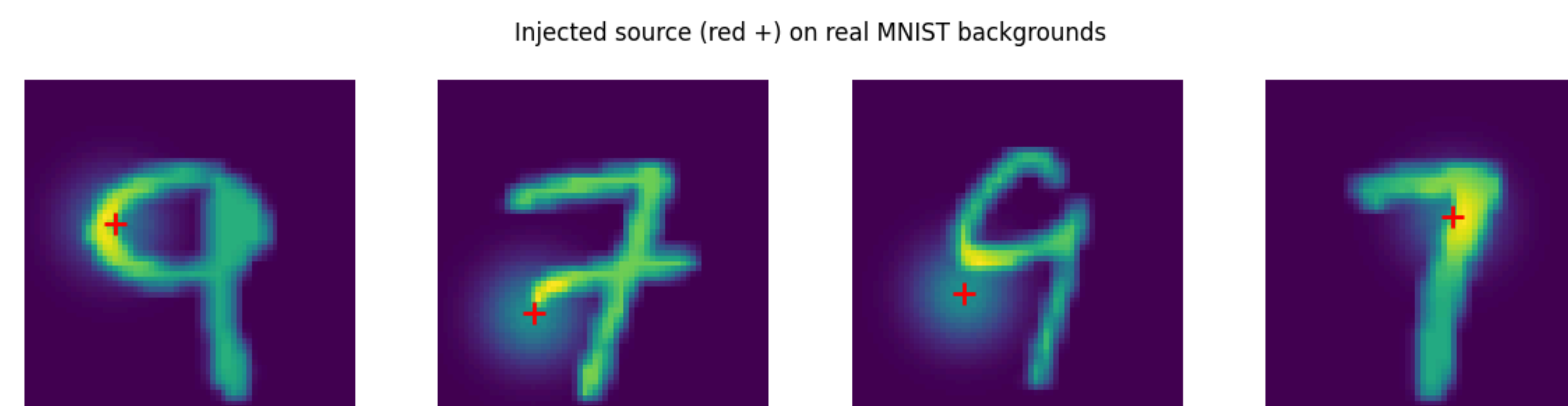
The SVD layer keeps the batch's **top-k singular directions** and normalizes them by their singular values:

$$X = U \Sigma V^T \rightarrow \hat{X} = \sqrt{N} \cdot U_k V_k^T \quad (\sigma_i \rightarrow 1)$$



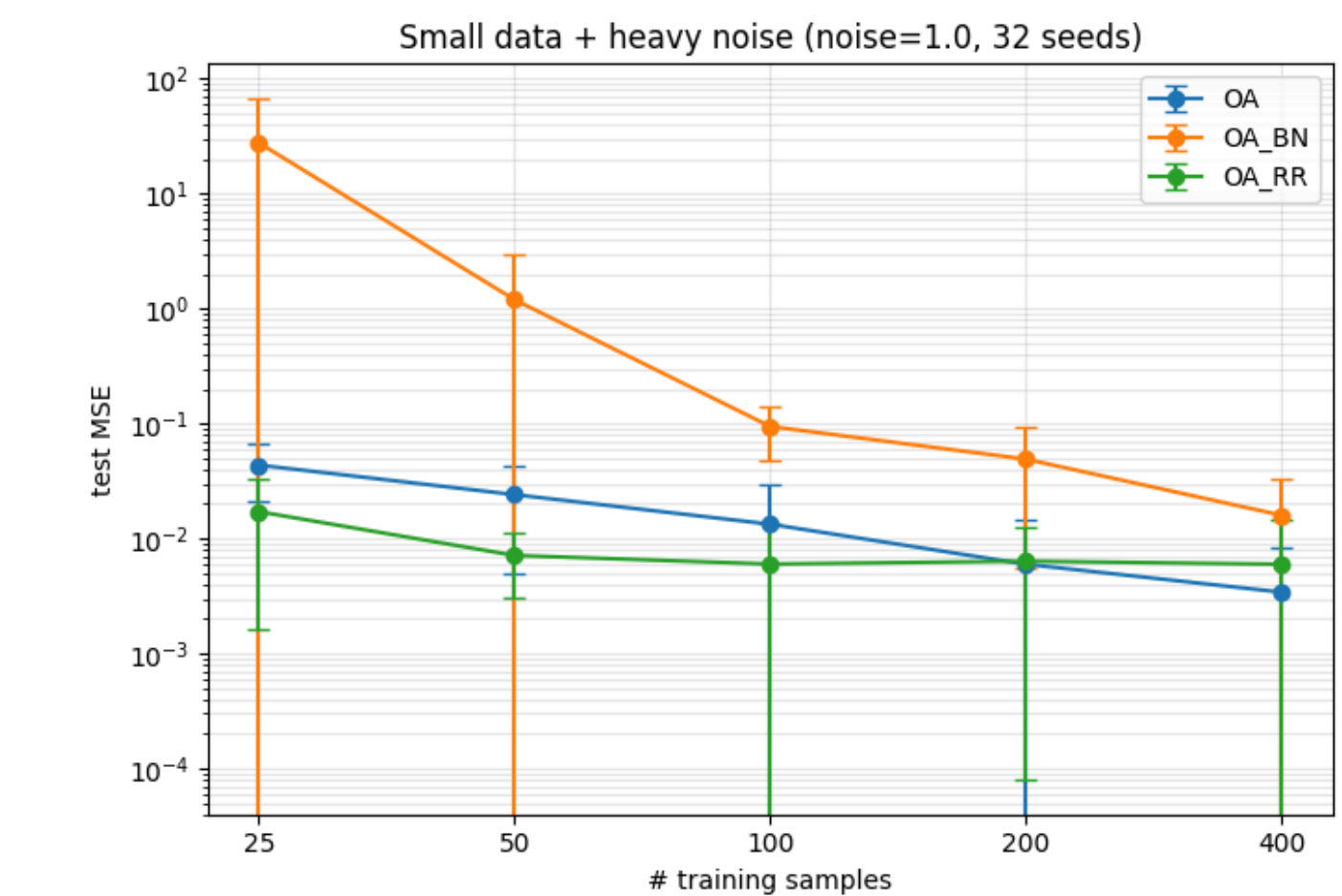
BN-style normalization, but in the batch's own decorrelated SVD basis.

- Task: regress a Gaussian source's centre (x, y) from a 64×64 image — **point-source localization**, as in astrometry and microscopy.
- Compare **Plain vs BN vs SVD** from **identical initial weights**, over up to 32 seeds — sweeping rank k, noise level and type, and dataset size N.
- External validity via **source injection**: a known source planted on real image backgrounds (astronomy's own validation method).

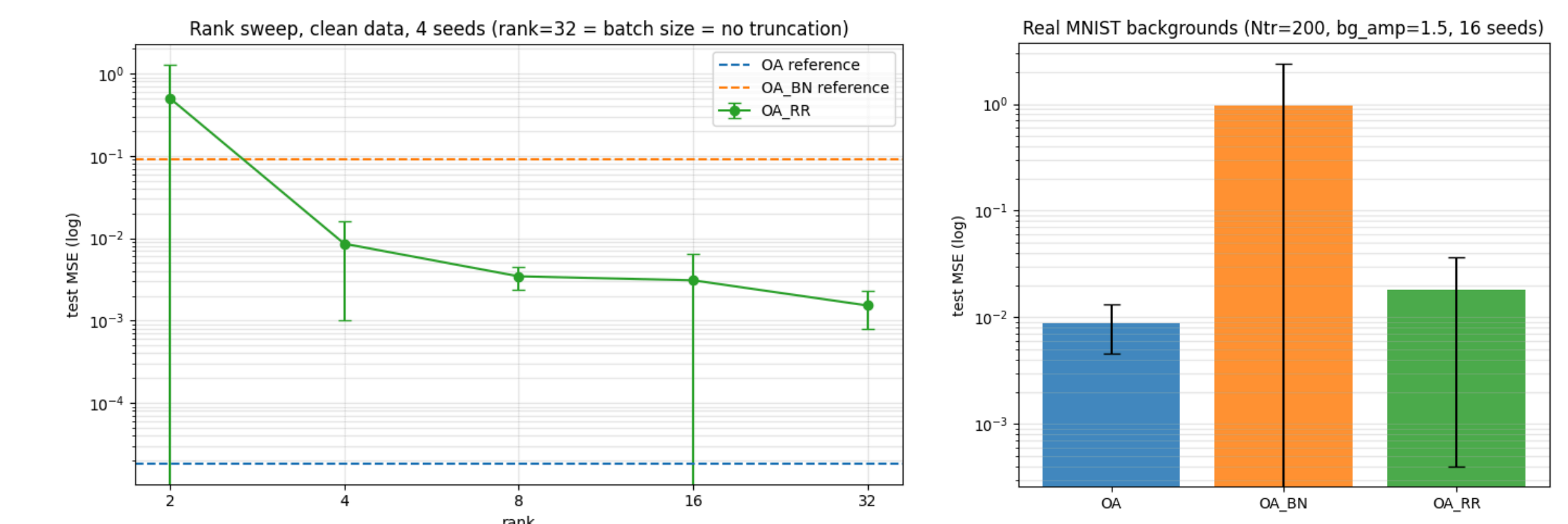


Injected source (red +) on real backgrounds: ground truth stays exact, clutter statistics become real.

04. Results



Small data + heavy noise (32 seeds): the SVD layer (green) beats BN everywhere and the plain net for $N \leq 100$. (OA = plain, OA_RR = SVD layer.)



Error falls monotonically with rank k.

Real backgrounds: SVD $\approx$ 50× better than BN.

- SVD beats BN in every like-for-like bottleneck comparison** — often by orders of magnitude — though per-layer BN on clean, plentiful data remains stronger.
- Beats the plain net **2–3× when data is scarce and noisy** ($N \leq 100$); ties at $N \approx 200$.
- Frozen inference basis is free (0.0028 vs 0.0034 MSE); **per-layer insertion diverges** — use one bottleneck.

05. Conclusion

- The SVD layer is a **more robust drop-in alternative to BN**, and outperforms even the unregularized network in the small-data, high-noise regime of source localization.
- Its one weakness: keeping top-k passes **structured low-rank nuisance** through — the inverse of astronomy's KLIP. A **reject-top-k** variant is the natural next step.

References

- J. Mounayer et al. *Rank Reduction Autoencoders*. arXiv:2405.13980, 2024.
S. Ioffe, C. Szegedy. *Batch Normalization*. ICML 2015.
R. Soummer et al. *Detection and Characterization of Exoplanets using Projections on Karhunen-Loève Eigenimages (KLIP)*. ApJL 2012.